General Guidelines

- **Clean Code:** Write your code in a clear, maintainable manner. Focus on proper naming conventions, code organization, and documentation.
- **UML Diagrams:** Ensure that your UML diagrams accurately represent the structure and relationships of your classes.
- **Professionalism:** Approach this project with a professional mindset. The quality of your work should reflect industry standards.

Resource Link

To assist you in completing these projects, I've provided a link to a guide that will help you understand the expectations and best practices for your work. Please refer to it as you progress: Project Guide Link

Final Remarks

Please don't hesitate to reach out if you have any questions or need further clarification. I'm looking forward to seeing the great work you will produce.

Library Management System-Project1

:Purpose and Concept

The Library Management System (LMS) is designed to automate the tasks involved in managing a library's book inventory and its members. The goal is to create a software system that makes it easy for librarians to track and manage books and for members to borrow and return them efficiently. This project serves as an introduction to fundamental OOP principles, such as encapsulation, .inheritance, and abstraction

Main Features:

Book Management -

Add Books:

.Create new entries for books, including details like title, author, ISBN, and publication date

Update Books : **-**

.Modify existing book details, such as correcting errors or updating availability status

Delete Books : **-**

.Remove books from the catalog when they are no longer available or needed

Member Management -

Register Members : -

Add new library members with personal details such as name, membership ID, and contact .information

Update Member Information : -

.Edit member profiles, including contact details and membership status

Track Borrowing History : -

.Maintain a record of all books borrowed by each member, including due dates and overdue fines

Borrowing and Returning Books

Borrow Books: Allow members to borrow books, updating their status to ‘checked out’ and - .recording the borrowing date

Return Books :Process the return of books, updating their status to ‘available’ and calculating .any overdue fines if applicable

UML Diagram Instructions

Create a UML Class Diagram with the following elements

Classes -

Book` : Attributes include title, author, ISBN, publication date, and status (available/checked ` - .out)

.Member` : Attributes include name, membership ID, contact information, and borrowed books` -

Library` : Methods include addBook(), removeBook(), registerMember(), borrowBook(), and ` - .()returnBook

Relationships:

Association :The `Library` class should have a one-to-many relationship with both `Book` and **-**
.``Member

.`**Aggregation** :A `Member` can have multiple borrowed `Books **-**

Online Shopping Cart System-Project 2

Purpose and Concept

The Online Shopping Cart System simulates an e-commerce platform where users can browse a catalog of products, manage their shopping carts, and proceed to checkout. This project explores advanced OOP concepts, including polymorphism, interfaces, and composition, and integrates .user interaction with product management and checkout processes

Main Features

Product Catalog -

Browse Products : -

.Display a list of products with details such as name, description, price, and stock quantity

Search and Filter: -

.Allow users to search for products and apply filters based on categories or price ranges

Shopping Cart -

Add/Remove Items: -

.Enable users to add products to their cart, update quantities, or remove items

View Cart:** -

. Provide a summary view of the cart, including item details, quantities, and total price

Apply Discounts: -
.Implement functionality to apply discount codes or promotions to the cart

Checkout Process -

Order Summary :Display a final summary of the order before purchase, including total cost and - .shipping details

Payment Processing:Simulate payment processing, which might include integrating with a - .mock payment gateway

Order Confirmation : Provide confirmation to users after a successful purchase, including order -
.details and estimated delivery time

UML Diagram Instructions

Create a UML Class Diagram with the following elements

Classes -

.**Product**` : Attributes include name, description, price, and stock quantity` -

.()**Cart**` : Methods include addItem(), removeItem(), updateQuantity(), and calculateTotal` -

.**User**` : Attributes include user information and cart details` -

.**Order**` : Attributes include order summary, payment status, and shipping details` -

Relationships -

.`**Aggregation**`User` has a `Cart`, and a `Cart` contains multiple `Products** -

.`**Association**`Order` is associated with `User` and contains a `Cart** -

Hotel Reservation System -Project3

Purpose and Concept:

The Hotel Reservation System manages the complex tasks associated with hotel bookings, including room availability, reservations, and check-ins/outs. This project involves implementing advanced OOP concepts and design patterns, focusing on the interaction between multiple .classes and handling data persistence

Main Features:

Room Management -

Room Types:Manage different types of rooms (e.g., single, double, suite) with varying rates and - .features

.**Availability**:Track room availability and update status based on reservations -

.**Room Details**:Include information such as room number, type, price per night, and amenities -

Booking System

.**Create Reservations**:Allow guests to book rooms for specific dates and track availability -

Modify Reservations:Enable changes to existing bookings, such as extending the stay or - .changing room types

.**Cancel Reservations:**Handle cancellation requests and update room availability accordingly -

Check-In/Check-Out -

.**Check-In Process**:Register guests upon arrival, assign rooms, and update room status -

.**Check-Out Process:**Process guest departures, finalize billing, and update room status -

Reporting -

Occupancy Reports: Generate reports on room occupancy rates, including peak times and - .average stay duration

.**Revenue Reports**:Track revenue generated from bookings and additional services -

UML Diagram Instructions

Create a UML Class Diagram with the following elements

Classes -

.Room` : Attributes include room number, type, status, and price` -

.Booking` : Attributes include reservation details, guest information, and booking dates` -

.Guest` : Attributes include personal information and reservation history` -

.()Hotel` : Methods include checkIn(), checkOut(), and manageRoomAvailability` -

Relationships -

.`**Association**:` Hotel` manages multiple ` Rooms` and ` Bookings -

.`**Aggregation**:` Booking` is associated with a ` Guest` and a ` Room -